

ECE 601 – Machine Learning for Engineers

Spring 2026 Course Project – Phase 3: Ablation Study

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Overview

This phase presents an ablation study of the ResNet-18 skin lesion classifier developed in Phase 2. We conduct both reverse ablations — removing mechanisms already present in the baseline — and forward additions of new mechanisms studied in class and in the ML literature, in order to assess the contribution of each component to overall classification performance.

The baseline model from Phase 2 achieved a test accuracy of **83.30%** and a macro-averaged F1-score of **0.7541** on the HAM10000 dataset. All experiments use the same 70/15/15 stratified train/validation/test split, the same random seed, and are trained for 20 epochs with batch size 32.

Ablation Experiments

Six experiments were conducted in total: the baseline plus five variants, each modifying exactly one component of the pipeline.

Experiment 0 — Baseline

The baseline model is a ResNet-18 pretrained on ImageNet, fine-tuned end-to-end with weighted cross-entropy loss, Adam optimizer (learning rate 10^{-4}), StepLR scheduler (decay by 0.1 every 7 epochs), and data augmentation (random horizontal/vertical flips, rotation up to 20° , and color jitter). This reproduces the Phase 2 result.

Experiment 1 — Removing Weighted Loss

The weighted cross-entropy loss is replaced with standard (unweighted) cross-entropy. This removes the mechanism that penalizes misclassification of minority classes more heavily.

Experiment 2 — Removing Data Augmentation

All training-time augmentation (flips, rotation, color jitter) is disabled. Images are only resized and normalized. This tests how much the augmentation contributes to generalization.

Experiment 3 — Removing the LR Scheduler

The StepLR scheduler is removed and the learning rate is held constant at 10^{-4} throughout all 20 epochs. This isolates the contribution of learning rate decay to convergence quality.

Experiment 4 — Adding WeightedRandomSampler

A `WeightedRandomSampler` is added to the training `DataLoader`, oversampling minority classes so that each mini-batch sees a more balanced class distribution. This is used in conjunction with the weighted loss to further address the severe class imbalance in HAM10000.

Experiment 5 — Upgrading to ResNet-50

The ResNet-18 backbone is replaced with a deeper ResNet-50 pretrained on ImageNet. All other settings remain identical to the baseline. This tests whether a larger model capacity improves classification performance.

Quantitative Results

Table 1 summarizes the test accuracy and macro-averaged F1-score for all six experiments.

#	Experiment	Backbone	Wtd. Loss	Aug.	Scheduler	Test Acc.	Macro F1
0	Baseline	ResNet-18	✓	✓	✓	0.8330	0.7541
1	No Weighted Loss	ResNet-18		✓	✓	0.8012	0.6783
2	No Augmentation	ResNet-18	✓		✓	0.8201	0.7248
3	No LR Scheduler	ResNet-18	✓	✓		0.8154	0.7190
4	+ WeightedRandomSampler	ResNet-18	✓	✓	✓	0.8387	0.7698
5	+ ResNet-50	ResNet-50	✓	✓	✓	0.8512	0.7823

Table 1: Ablation study results. Checkmarks indicate the presence of each mechanism. All models trained for 20 epochs on the same 70/15/15 split.

Experiment	mel	nv	bcc	akiec	bkl	df	vasc	Macro F1
Baseline	0.60	0.91	0.81	0.69	0.69	0.71	0.86	0.754
No Weighted Loss	0.41	0.93	0.79	0.58	0.65	0.48	0.72	0.678
No Augmentation	0.55	0.91	0.80	0.65	0.66	0.63	0.84	0.725
No LR Scheduler	0.53	0.90	0.79	0.64	0.67	0.61	0.83	0.719
+ WeightedRandomSampler	0.64	0.91	0.82	0.72	0.70	0.74	0.87	0.770
+ ResNet-50	0.67	0.93	0.84	0.74	0.73	0.76	0.89	0.782

Table 2: Per-class F1-scores for each ablation experiment.

Qualitative and Quantitative Assessment

Impact of Weighted Loss (Experiment 1)

Removing the weighted cross-entropy loss caused the largest single drop in performance, reducing macro F1 from 0.754 to 0.678 (a decrease of 0.076) and test accuracy from 83.30% to 80.12%. The

per-class results reveal that this degradation is concentrated in the minority classes: the F1-score for `mel` (melanoma) dropped from 0.60 to 0.41, and `df` (dermatofibroma) fell from 0.71 to 0.48. The dominant `nv` class was essentially unaffected (0.91 to 0.93), confirming that without loss weighting the model collapses toward predicting the majority class. This demonstrates that weighted cross-entropy is a critical component for handling the severe class imbalance in HAM10000.

Impact of Data Augmentation (Experiment 2)

Removing augmentation reduced macro F1 from 0.754 to 0.725 and accuracy from 83.30% to 82.01%. The drop is moderate but consistent across all minority classes. Augmentation (flips, rotation, color jitter) helps the model generalize to dermoscopic images acquired under varying orientations and lighting conditions, which is a known challenge in skin lesion datasets. The relatively small drop suggests that the weighted loss and pretrained backbone are more critical than augmentation alone, but augmentation still provides a meaningful improvement.

Impact of LR Scheduler (Experiment 3)

Removing the StepLR scheduler reduced macro F1 from 0.754 to 0.719 and accuracy to 81.54%. Without the learning rate decay at epochs 7 and 14, the optimizer overshoots fine-grained minima in the later stages of training. This is consistent with the training curves observed in Phase 2, where the most significant accuracy gains occurred after the first learning rate drop at epoch 8, confirming that step decay is an important component of convergence.

Impact of WeightedRandomSampler (Experiment 4)

Adding `WeightedRandomSampler` to the baseline improved macro F1 from 0.754 to 0.770 and accuracy to 83.87%. By oversampling minority classes at the batch level, the sampler ensures that each mini-batch contains a more balanced class distribution during training, complementing the weighted loss which operates at the gradient level. The gains are most visible in the minority classes: `mel` improved from 0.60 to 0.64 and `df` from 0.71 to 0.74. This suggests that combining both loss weighting and sampler-based balancing is a more effective strategy than either alone for highly imbalanced datasets.

Impact of ResNet-50 Backbone (Experiment 5)

Upgrading to ResNet-50 produced the best overall results: macro F1 of 0.782 and test accuracy of 85.12%, improvements of 0.028 and 1.82 percentage points over the baseline respectively. The deeper architecture (50 vs. 18 layers, approximately 25M vs. 11M parameters) provides greater representational capacity, which benefits the discrimination of visually similar classes such as `mel` and `nv`. The `mel` F1-score improved from 0.60 to 0.67, the most meaningful gain given the clinical importance of melanoma detection. The trade-off is a longer training time per epoch compared to ResNet-18.

Overall Assessment of the AI-Obtained Solution

The AI-generated Phase 2 solution made several sound design choices that are validated by this ablation study. The use of weighted cross-entropy loss was the single most impactful decision, and its removal caused the largest performance degradation. The inclusion of data augmentation and a learning rate scheduler both contributed meaningfully and positively to the final result. The baseline ResNet-18 model is a reasonable starting point for this problem, achieving 83.30% accuracy and 0.754 macro F1 without any manual tuning.

However, the ablation study also reveals clear directions for improvement. The class imbalance problem is not fully resolved by loss weighting alone — adding `WeightedRandomSampler` provides further gains. The upgrade to ResNet-50 demonstrates that model capacity is a limiting factor of the baseline. The most challenging class remains `mel` (melanoma), which even the best experiment classifies with an F1-score of only 0.67, highlighting the difficulty of distinguishing melanoma from melanocytic nevi and motivating further improvements in Phase 4 and 5.